المدرسة الوطنية للعلوم التطبيقية - بني ملال

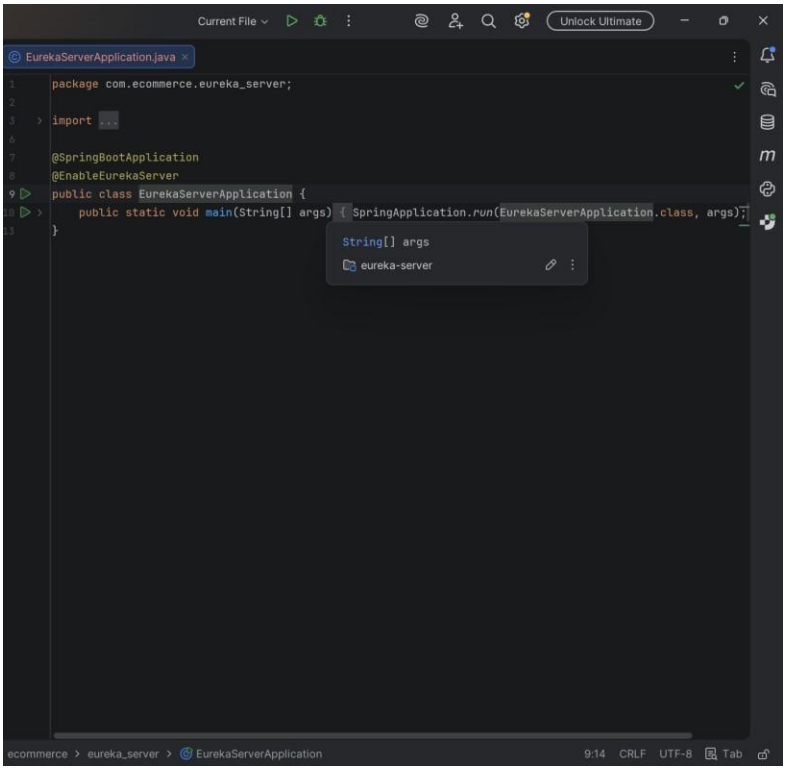
ⵜⴰⵙⴷⴰⵡⵉⵜ ⵏ ⵜⵎⴰⵙⵙⴰⵏⵉⵏ ⵜⵓⵏⵉⵏⵉⵏ - ⴱⵏⵉ ⵎⵍⴰⵍ

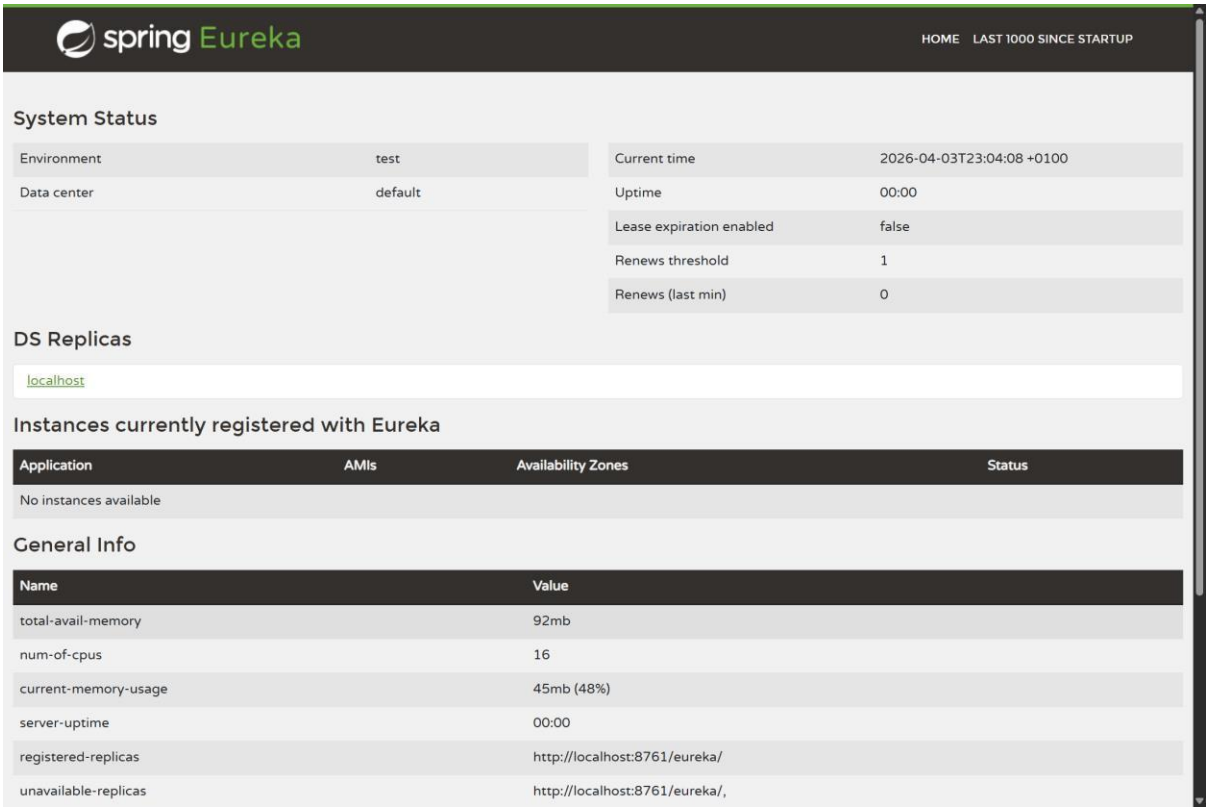
Ecole Nationale des Sciences Appliquées - Béni Mellal

- **Nom de l'étudiant :** Adam El Mansour
- **Encadre par** : Pr : B.ELBAGHAZAOUI

TP4 : Architecture Orientée Service (SOA)

Etape 1 :


Current File
Unlock Ultimate
EurekaServerApplication.java
package com.ecommerce.eureka_server;
import ...
@SpringBootApplication
@EnableEurekaServer
public class EurekaServerApplication {
public static void main(String[] args) { SpringApplication.run(EurekaServerApplication.class, args);
}
String[] args
eureka-server
ecommerce > eureka_server > EurekaServerApplication
9:14 CRLF UTF-8 Tab


spring Eureka
HOME LAST 1000 SINCE STARTUP
System Status
Environment test
Data center default
Current time 2026-04-03T23:04:08 +0100
Uptime 00:00
Lease expiration enabled false
Renews threshold 1
Renews (last min) 0
DS Replicas
localhost
Instances currently registered with Eureka
Application AMIs Availability Zones Status
No instances available
General Info
Name Value
total-avail-memory 92mb
num-of-cpus 16
current-memory-usage 45mb (48%)
server-uptime 00:00
registered-replicas http://localhost:8761/eureka/
unavailable-replicas http://localhost:8761/eureka/,

Etape 2 :

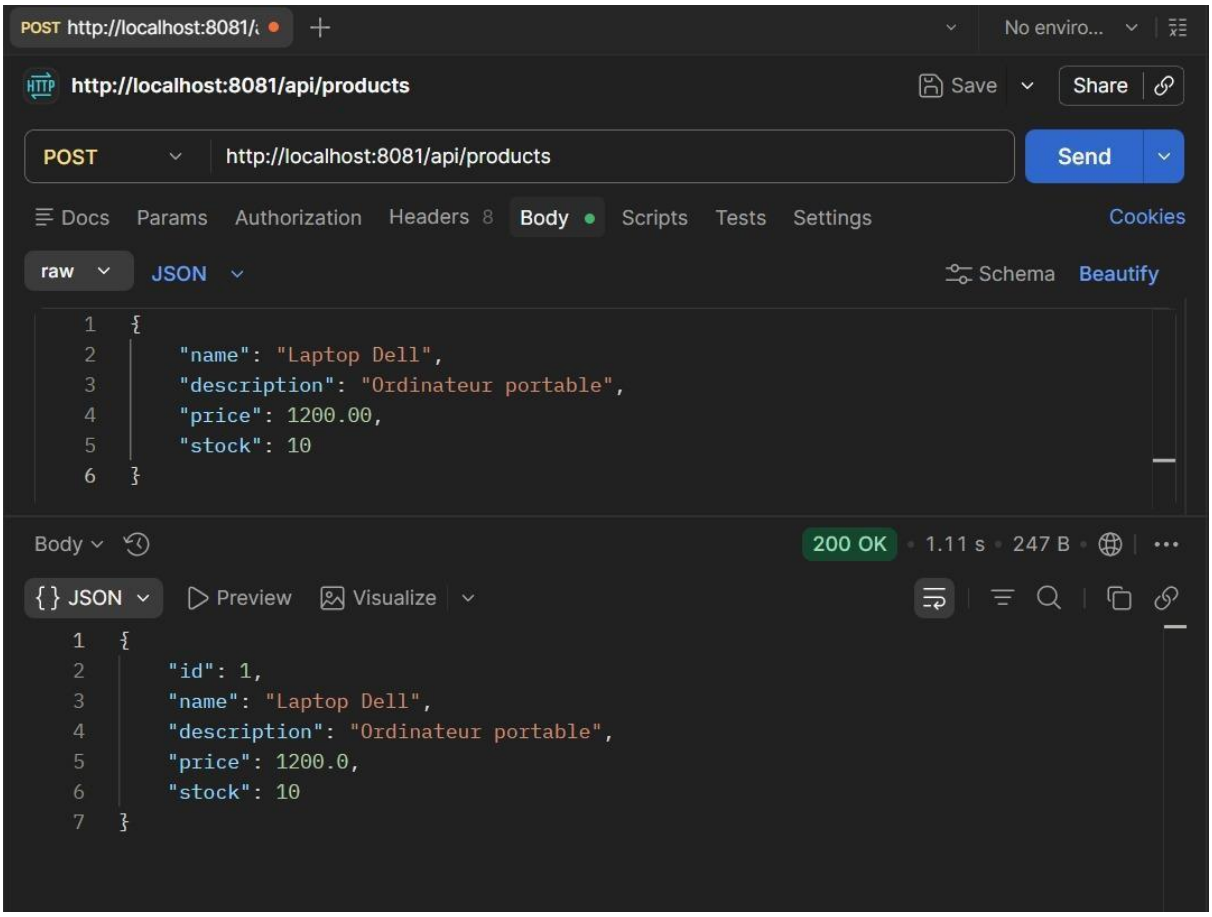
```java
package com.ecommerce.product_service.service;

import ...

@Service
@RequiredArgsConstructor
public class ProductService {
    private final ProductRepository productRepository;

    public List<Product> getAllProducts() { return productRepository.findAll(); }
    public Optional<Product> getProductById(Long id) { return productRepository.findById(id); }
    public Product createProduct(Product product) { return productRepository.save(product); }
    public void deleteProduct(Long id) { productRepository.deleteById(id); }

    public Product updateStock(Long id, Integer quantity) {
        Product product = productRepository.findById(id)
                .orElseThrow(() -> new RuntimeException("Produit non trouvé"));
        if (product.getStock() < quantity)
            throw new RuntimeException("Stock insuffisant");
        product.setStock(product.getStock() - quantity);
        return productRepository.save(product);
    }
}
```

Etape 3 :

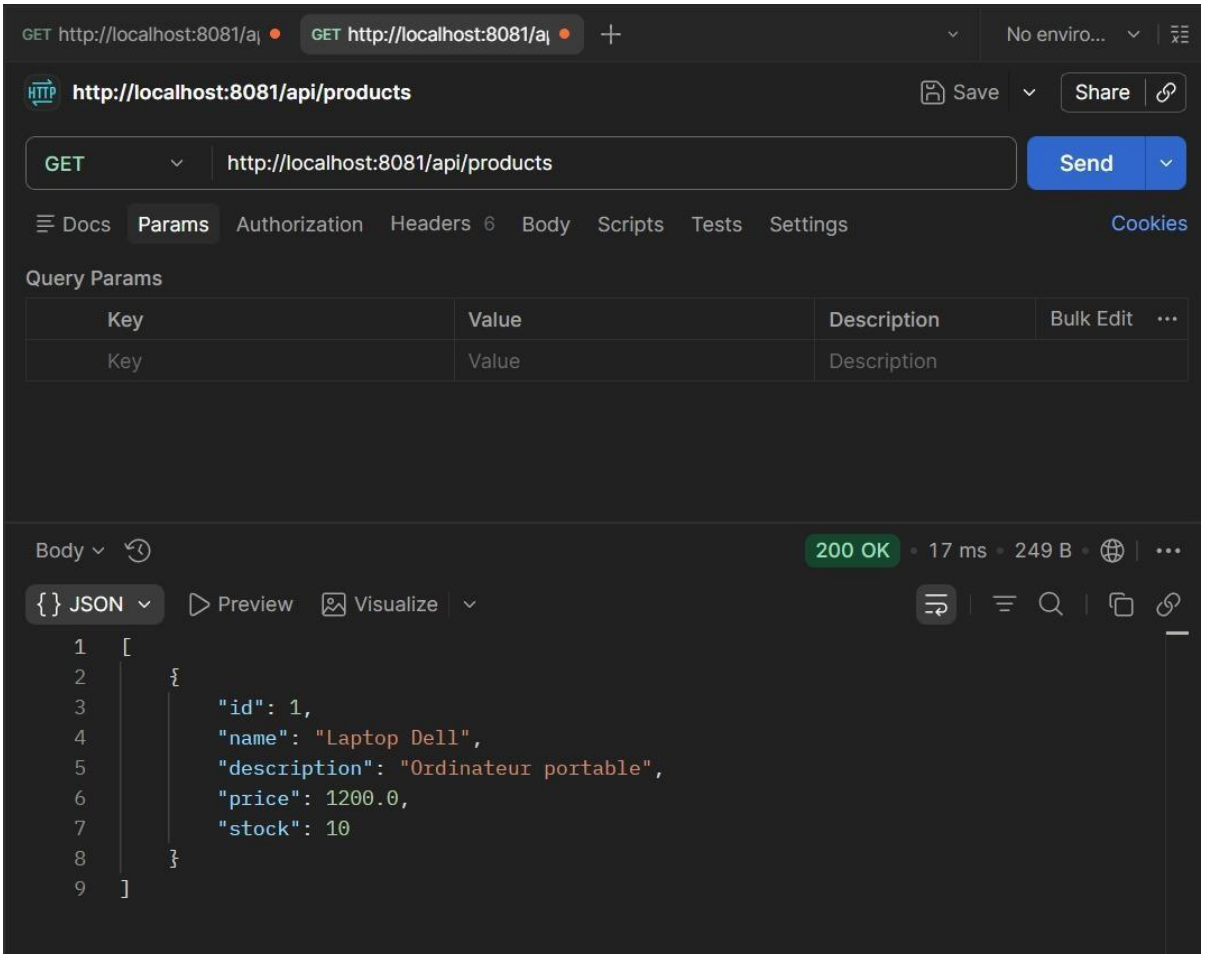
```java
package com.ecommerce.order_service.service;

import ...

@Service
@RequiredArgsConstructor
public class OrderService {
    private final OrderRepository orderRepository;
    private final ProductClient productClient;

    public List<Order> getAllOrders() { return orderRepository.findAll(); }

    public Order createOrder(Long productId, Integer quantity) {
        ProductDTO product = productClient.getProductById(productId);
        if (product.getStock() < quantity)
            throw new RuntimeException("Stock insuffisant");
        Order order = new Order();
        order.setProductId(productId);
        order.setProductName(product.getName());
        order.setQuantity(quantity);
        order.setTotalPrice(product.getPrice() * quantity);
        order.setOrderDate(LocalDateTime.now());
        order.setStatus("PENDING");
        productClient.updateStock(productId, quantity);
        return orderRepository.save(order);
    }
}
```

Tests :

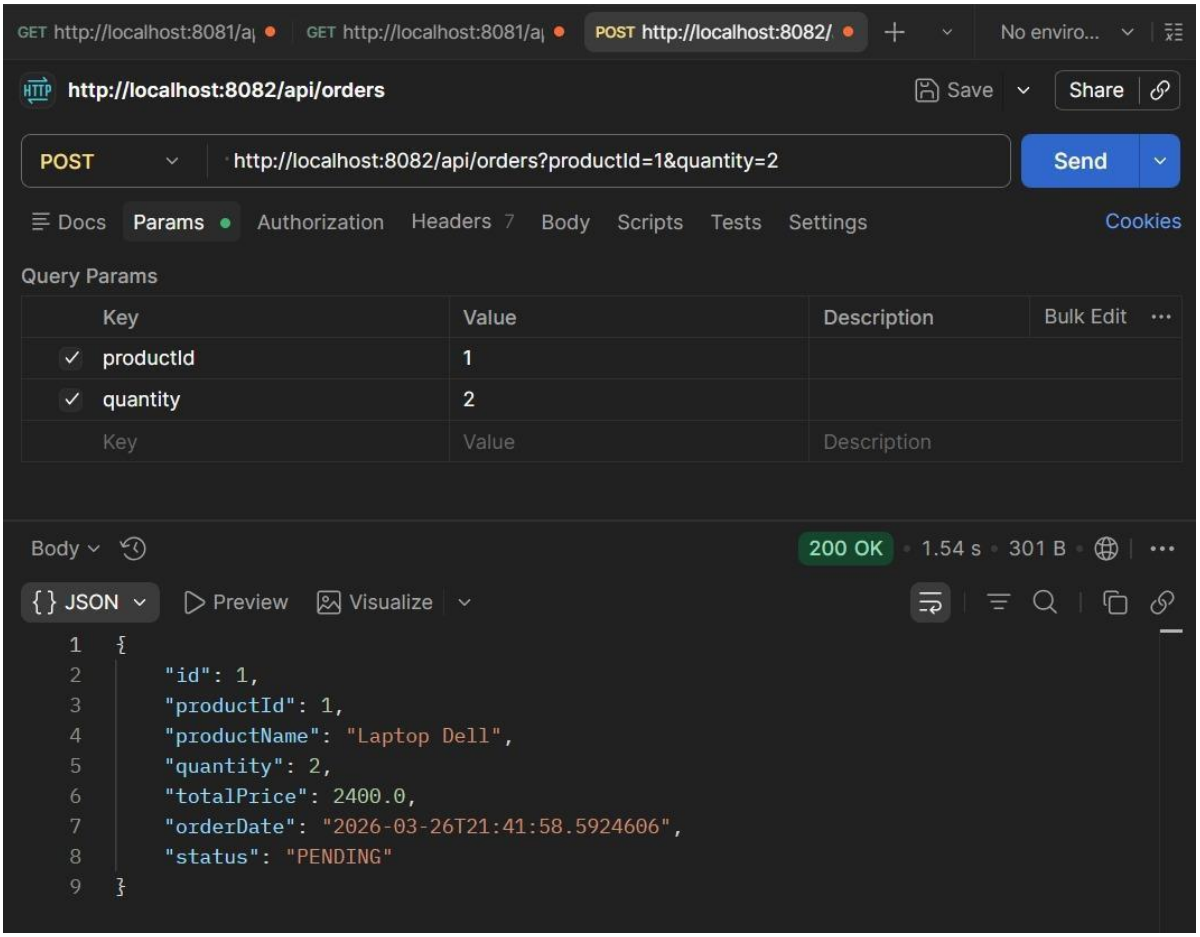
Test 1 :



Test 2 :


GET http://localhost:8081/a
GET http://localhost:8081/a
No enviro...
http://localhost:8081/api/products
Save
Share
GET
http://localhost:8081/api/products
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Tests
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Key
Value
Description
Body
200 OK
17 ms
249 B
JSON
Preview
Visualize
[
{
"id": 1,
"name": "Laptop Dell",
"description": "Ordinateur portable",
"price": 1200.0,
"stock": 10
}
]

Test 3 :


GET http://localhost:8081/a
GET http://localhost:8081/a
POST http://localhost:8082/
No enviro...
http://localhost:8082/api/orders
Save
Share
POST
http://localhost:8082/api/orders?productId=1&quantity=2
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Tests
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
productId
1
quantity
2
Key
Value
Description
Body
200 OK
1.54 s
301 B
JSON
Preview
Visualize
{
"id": 1,
"productId": 1,
"productName": "Laptop Dell",
"quantity": 2,
"totalPrice": 2400.0,
"orderDate": "2026-03-26T21:41:58.5924606",
"status": "PENDING"
}

Test 4 :

GET http://localhost | GET http://localhost | POST http://localhos | GET http://localhost | No enviro...

http://localhost:8082/api/orders

Save | Share

GET http://localhost:8082/api/orders Send

Docs | Params | Authorization | Headers 6 | Body | Scripts | Tests | Settings | Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body | 200 OK • 567 ms • 302 B

{ } JSON | Preview | Visualize


```
[
    {
        "id": 1,
        "productId": 1,
        "productName": "Laptop Dell",
        "quantity": 2,
        "totalPrice": 2400.0,
        "orderDate": "2026-03-26T21:41:58.592461",
        "status": "PENDING"
    }
]
```